

the chips need to be tested in several different environments. Chips are tested for various domains like electrical validation, thermal characterisation, voltage and current graphs.

Electrical design validation helps to assess whether the given IC complies with a variety of different operating specifications, while reliability analysis helps to deduce the long-term reliability of the semiconductor and its suitability for intended applications.

Veerappan explains, "If a chip has a certain operating temperature, its performance is tested for varying temperature. Similarly, if the chip operates at 2 GHz, its operation is tested at 1.9 GHz and 2.1 GHz." It's like quality certification of the finished device before it goes for mass packaging.

Kumar says, "In addition to functional testing, DC and AC parametric testing is a big challenge, as with new processes the transistors need to be characterised and verified for their DC and AC characteristics. There are parametric tests which are performed to understand the semiconductor reliability, lifetime, device breakdown limits, etc. Typically, they involve current-voltage and capacitance-voltage characterisation of devices and then deriving reliability and other parameters from these tests."

Biggest claim: chip is not working

The first working chip itself is a big challenge. Varambally says, "Another big challenge is to conclude that chip is faulty. One person cannot comment on that. So you need a team of experts to conclude that the chip is faulty or the design is faulty."

Detecting a bug is not the only key thing in testing, pinpointing it in a large and complicated chip is also very important. "Several years ago, designing and testing were two independent processes. Today, a new process of concurrent or co-design and verification is used where the designer and

Equipment for post-silicon testing

- Automated test equipment (ATEs) or silicon testers
- Logic analysers
- Pulse pattern generators
- Arbitrary waveform generators
- Oscilloscopes
- Digitisers
- Semiconductor parametric analysers
- EDA tools
- Design-for-testability tools

test engineer work together to detect the fault at an early stage," shares Neeraj Bhardwaj, systems hardware architect, Texas Instruments.

If the chip fails, a series of tests are performed to locate the fault. This is done by the chip designer himself. Veerappan says, "Sometimes we cut open the chip to find the problem and inform the designer where the design went wrong."

Logical tests help in detecting where the input and output mismatch. "These digital test results help to find the point of failure and do detailed analysis," explains Varambally. He adds that checking the nodes of abnormal current and voltage, waveform shapes and signals and comparing them with the expected values helps in debugging.

Even standard test equipment like logic analysers and pattern generators have evolved over the years to help ensure that there is a smooth handover from design simulation to actual device test. Kumar explains, "This is achieved by providing linkage for accepting test vector inputs from design tools. Designers can create test vectors in VHDL, etc, and the same can be imported into pattern generators to actually generate and apply the same to the ICs as digital stimulus."

Design for testability

You must be able to test every design to ensure the intent of the design. "If the design is not testable, it may be very difficult to find the fault after

packaging. This involves cost and time," explains Veerappan. There are different software tools available from companies like Cadence and Synopsys to design for testability.

Varambally explains, "Let's say there are millions of transistors, but only hundreds of pins. Practically, you should be able to test any chip failing through only these pins. The concept of design for testability applies to testing of all the transistors or different sections of the chip." There are different test nodes created to test various sections of the IC.

Advanced test equipment are very expensive. So the time involved in testing each chip is very critical. "Design for test helps reduce the time to test one chip, thereby increasing the utilisation factor of the tester. So you need to design the chip such that your testing time comes down," explains Varambally.

The cost of a chip design is millions of dollars, which also includes the cost of test. To minimise the cost and perform the test efficiently, sometimes an extra circuit is added to perform tests.

Varambally says, "We inculcate all the testing features in the design stage itself. For example, we can put a small extra structure along with the circuit, like a small block. Although there is a small price associated with this extra piece of silicon, it will reduce the testing time and cost. So it is okay to pay a bit of silicon fully for testing."

Embedded selftest inside the chip itself is a logical and viable future solution for complex SoC designs that saves test development cost and reduces the load board and ATE system complexity.

Sabapathi S.R., chief executive officer, Qmax, says, "The built-in selftest can be commanded by simple low-cost JTAG interface. This will cover the functionality of the device. Henceforth ATE test will be much simpler and in some cases limited to DC and AC parametric tests alone. The resultant savings easily justify additional silicon block embedded into the chip."